

 7. GOVERNANCE & RELEASE GATE · STPA Hazard Mitigation ·
Bumpless Override · Defensible Release Case (**LOOP-01** → **REL-01**)

 **1. SENSING**
Photons / Voltages to DMA
MIPI CSI-2 · SPI Bus Priority

 **2. PERCEPTION**
ViT Encoders · DINOv2
3D Spatial Affordance Tokens

 **3. MEMORY**
Temporal Belief · $SE(3)$ Trees
JEPA / RSSM World Models

 **5. PLANNING (System 1.5 · MPU/NPU)**
Diffusion Policies · ACT Action Chunking
 H -Step Action Horizons · Jerk Continuity

 **4. REASONING (System 2 · MPU)**
Vision-Language Foundation Models · 3D Goals
Leases with Expiring TTL (t_{expire})

 **THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY**

 **6. REFLEX (System 1 · MCU)**
1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}}$ · Veto (u_t)

 **THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)** · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

 Endogenous Sensory Shift (O_{t+1})